

Jonathan Jaime Luzuriaga Vélez

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PROFILE

Software Engineer (Systems & Applied AI) focused on developing resilient architectures and deploying AI models under hardware constraints (Edge Computing). Specialist in auditing systems from first principles to mitigate design vulnerabilities. Proven track record in legacy code refactoring, DevSecOps practices, and documenting complex technical reasoning to ensure data integrity and system scalability.

EDUCATION

Escuela Politécnica Nacional, Quito | Jun 2021 — Present | Software Engineering

- **Interests:** Logical optimization, Distributed architectures, Edge AI, Information security.
- **Thesis (In progress):** Emerging optimization paradigms in Deep Learning.

PROFESSIONAL EXPERIENCE

Software Engineer (Backend & Automation) | Manticore Labs (Remote) | Mar 2026 — Present

- Architect backend automation solutions, identifying and troubleshooting operational frictions between functional teams by redesigning logical workflows.
- Orchestrate complex data pipelines using n8n for centralized synchronization of corporate documentation via the Notion API.

Product Owner & Tech Lead | Lynx Cores (Vixora Tech) | Jan 2026 — Present

- **Award:** 2nd place at the Hult Prize EPN, achieving national qualification based on technical architectural feasibility.
- Lead a core engineering team in the development of an *offline-first* AgroTech application based on Edge Computing for crop disease detection, operating under strict latency and intermittent connectivity.

Remote Intern (ERP Development) | SERINDPET | Oct 2025 — Jan 2026

- Refactored relational database schemas in SQL Server through index optimization, eliminating performance bottlenecks in the concurrent generation of executive reports.
- Designed a scalable backend architecture using Next.js and Express, implementing secure deployment automation via CI/CD pipelines in GitHub Actions and Azure DevOps.

Backend Developer | Banco de Alimentos de Quito | Jun 2024 — May 2025

- Built and containerized microservices (Docker) on AWS for massive behavioral data ingestion, ensuring high availability across the infrastructure.

Full Stack Developer | Instituto Geográfico Militar | Apr 2024 — Dec 2024

- Integrated massive data flows between modern interfaces and institutional databases, modernizing legacy systems and applying strict government standards for encryption and sensitive data handling.

OPEN SOURCE AND INDEPENDENT PROJECTS

High-Security Architecture: The Secure Judge (El Juez Seguro)

- Designed an independent backend ecosystem applying strict Zero-Trust security principles and process isolation.
- Implemented cryptographic signature schemes to guarantee the immutability and integrity of legal records.
- Structured the repository under a monorepo model to centralize version control, dependency management, and facilitate scalable, secure deployments.

BirdCLEF+ 2026 (Cornell Lab of Ornithology) | Kaggle Competition

- Developed a multi-label classification pipeline to identify continuous wildlife acoustic events utilizing deep learning frameworks.
- Implemented signal processing techniques to isolate target vocalizations and mitigate ambient noise in complex wetland ecosystems.

SCIENTIFIC AND ACADEMIC PUBLICATIONS

A Unified Taxonomy of Deep Learning Optimizers for Scalable and Efficient AI Systems | *AHFE Open Access Scientific Paper*, vol 203 (*AHFE International*) | July 2026 | **DOI:** 10.54941/ahfe1008074

- Co-authored a structured decision framework classifying optimization algorithms by integrating IO-aware and Flash attention paradigms.
- Research focused on cost efficiency, memory usage, and hardware constraints to democratize the training and deployment of large-scale LLMs.

Risk Prediction Model (Smart Irrigation) | Python, TensorFlow, Scikit-learn

- Outperformed traditional climate models by implementing XGBoost and LSTM for satellite data analysis, utilizing Dask for out-of-core processing of high-volume datasets.

HACKATHONS AND CHALLENGES

Global Game Jam 2026 | Lead Developer (Unity, C#)

- I led the technical development of a 2D puzzle game in 48 hours, managing version control (Git) and resolving integration conflicts within a multidisciplinary team.

LICENSES AND CERTIFICATIONS

DevSecOps Essentials (DSE) – EC-Council | (Jul 2026 - Jul 2029) | **Credential:** ECC6947035182

- Official certification validating fundamentals in security practices seamlessly integrated into the continuous development lifecycle (Shift-Left Security).

SKILLS

- **Architecture & Backend:** Node.js, Express, Go, Zero-Trust Architectures, Monorepos, C4 Modeling.
- **Cloud & DevOps:** Docker, AWS, Azure DevOps, CI/CD (GitHub Actions), Firebase/Supabase.
- **AI & Data:** TensorFlow, PyTorch, Scikit-learn, Audio Event Classification, Edge AI, ETL concepts.
- **Languages & Databases:** Python, TypeScript/JavaScript, SQL (SQL Server, PostgreSQL), Java.
- **Quality & Methodologies:** Behavior-Driven Development (BDD), Gherkin, Agile, Git.
- **Languages:** Spanish (Native), English (Advanced).